

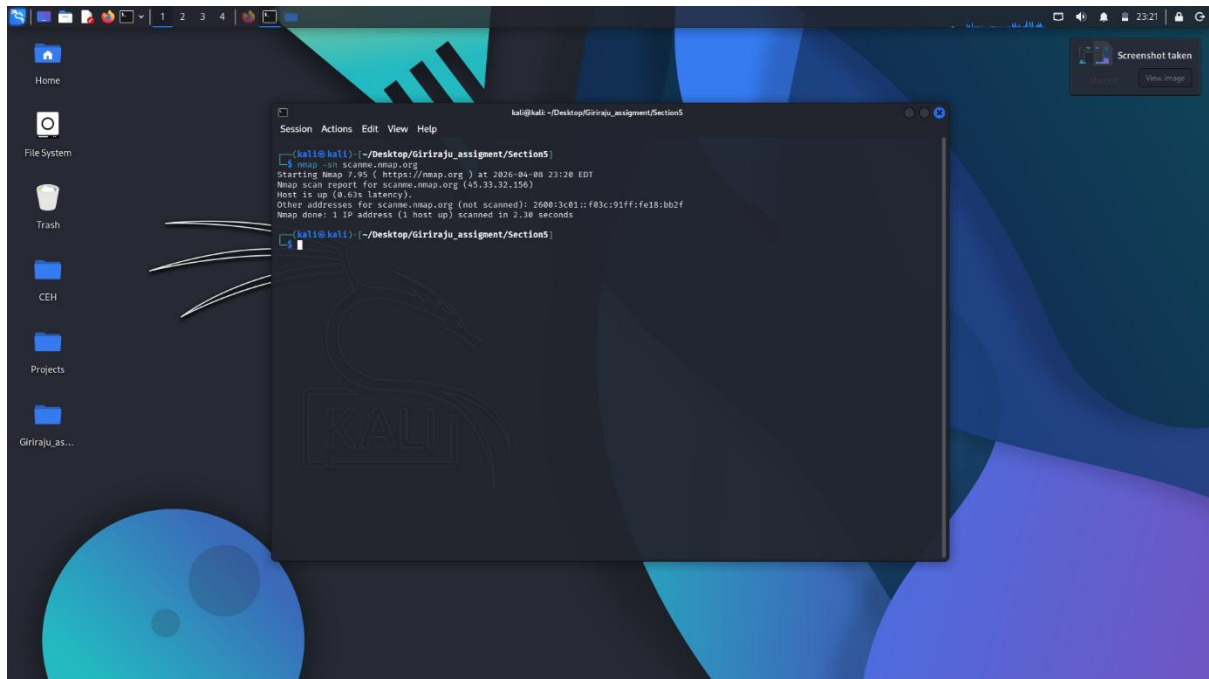
1. Ping Scan

Command:

```
nmap -sn scanme.nmap.org
```

Explanation:

This scan checks whether the target system is online. It does not scan ports, only verifies host availability.



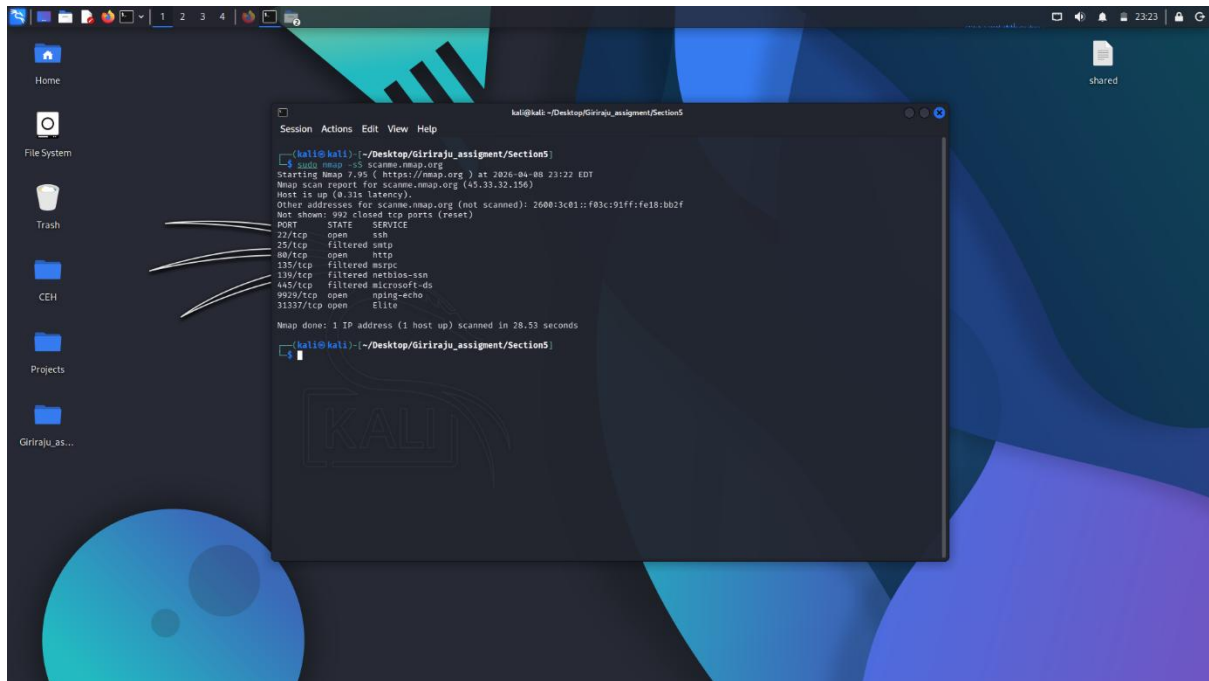
2. TCP SYN Scan

Command:

```
sudo nmap -sS scanme.nmap.org
```

Explanation:

This is a stealth scan that sends SYN packets to detect open ports without completing the full connection.



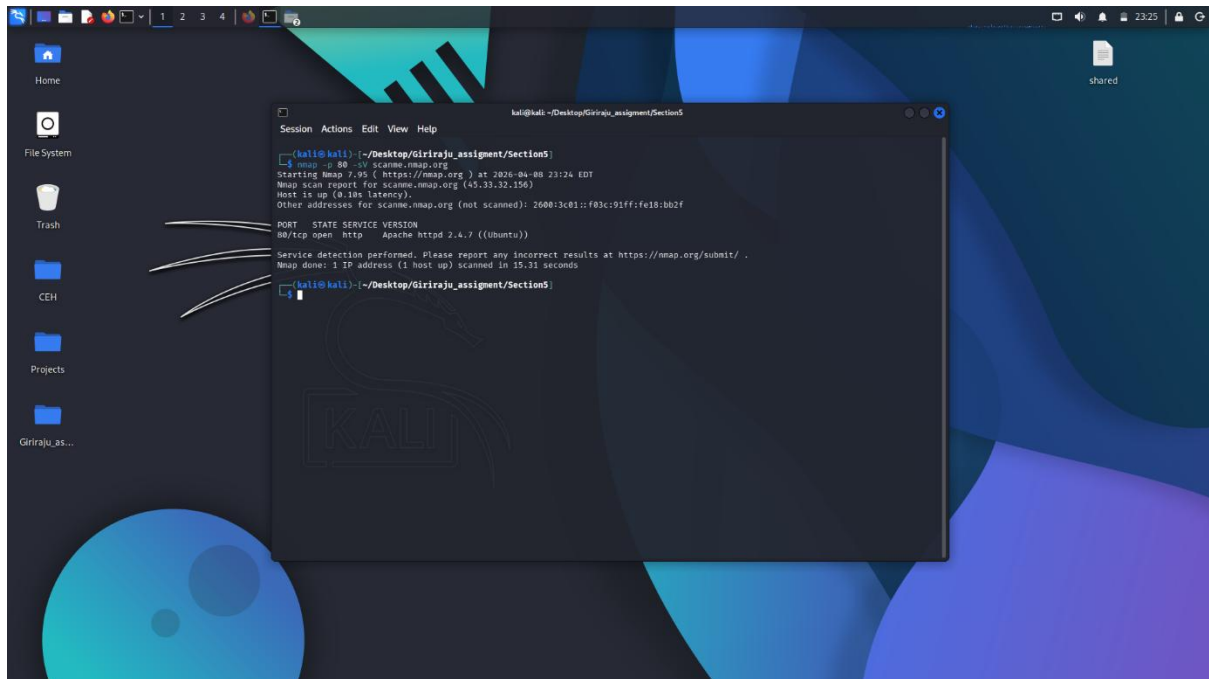
3. Service Version Detection

Command:

```
nmap -p 80 -sV scanme.nmap.org
```

Explanation:

This scan identifies the service and version running on a specific port (port 80 in this case).



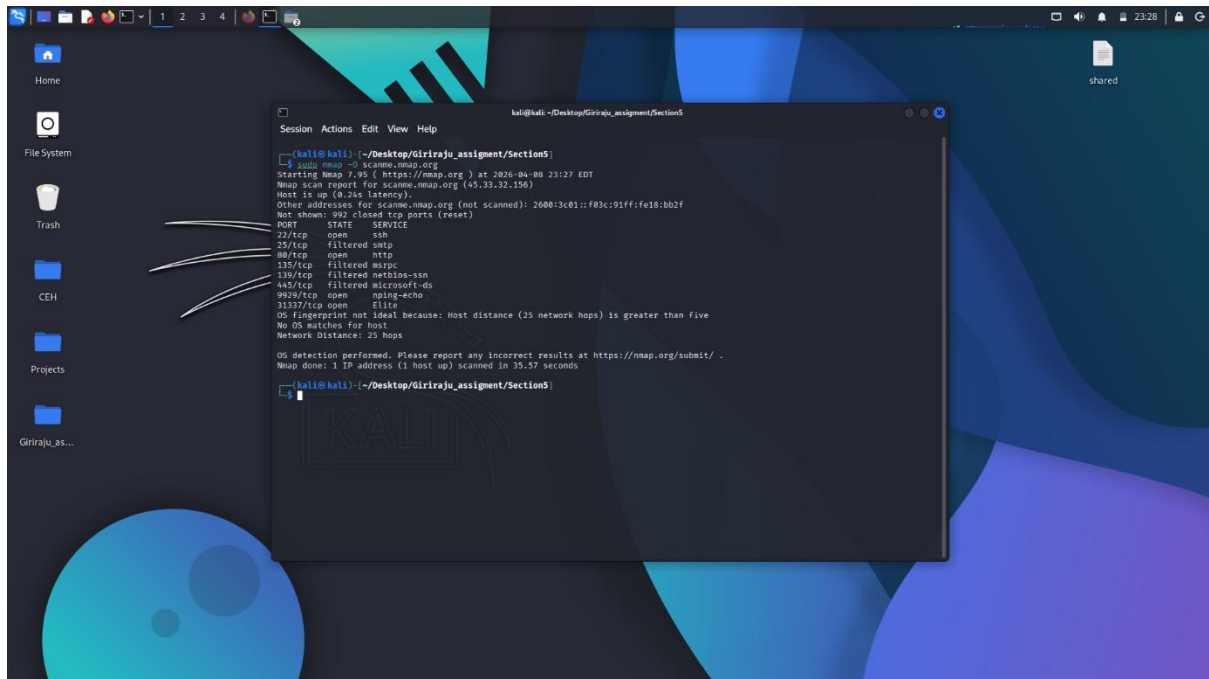
4. OS Detection

Command:

```
sudo nmap -O scanme.nmap.org
```

Explanation:

This scan attempts to detect the operating system of the target based on network behavior.



5. NSE Script Scan

Command:

```
nmap -sC scanme.nmap.org
```

Explanation:

This scan runs default Nmap scripts to gather additional information like vulnerabilities and services.

